

Leveraging OpenAI Whisper Model to Improve Speech Recognition for Dysarthric Individuals

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Abstract—Automatic Speech Recognition (ASR) systems are pivotal in facilitating human-technology interactions through voice commands. However, individuals with dysarthria face significant challenges in benefiting from these technologies due to their speech disorder. This paper proposes finetuning the Whisper model for Dysarthric Speech Recognition (DSR) by incorporating additional features extracted from Mel-frequency cepstral coefficients (MFCCs). By combining spectrograms and MFCCs within an attention mechanism, the model creates a richer feature representation, with spectrograms providing broader context and MFCCs highlighting crucial formant frequencies. The attention mechanism dynamically weighs the importance of each feature based on specific speech segments and dysarthric speech characteristics. Furthermore, a hierarchical attention approach is adopted, which encompasses a two-stage attention mechanism. This mechanism directs attention at both local and global levels, facilitating the capture of both fine-grained details and broader contextual information within the speech signal. This study involved the development and training of 45 speaker-adaptive dysarthric ASR systems. The proposed model achieves an average Word Recognition Accuracy (WRA) of 74.08%, showing a notable enhancement compared to the benchmark of 69.23%. The findings underscore the efficacy of the proposed approach in addressing dysarthria-related challenges in ASR systems.

Keywords—DSR, MFCC, UASpeech, WRA, Whisper

I. INTRODUCTION

In today's digital era, ASR technology serves a crucial role, facilitating seamless interaction with various devices such as smartphones and smart home systems [1]. ASR accurately transcribes spoken words, serving as an automated assistant in numerous applications. However, individuals with dysarthria encounter obstacles in fully utilizing ASR technology due to the complexities inherent in their speech patterns. Dysarthria, a neurological condition affecting the control of motor speech articulators, results in difficulties coordinating or paralysis of speech-related muscles. This impairment often leads to irregular speech patterns and reduced clarity, posing challenges for effective communication. Dysarthria notably changes speech, particularly impacting pronounced phonemes, especially in individuals with severe disabilities. These pronounced phonemes become indefinite, leading to inaccuracies in pitch pauses during the articulation of vowels and consonants [2]. Such discrepancies obscure the distinctive acoustic features crucial for ASR systems to differentiate between phonemes. Moreover, due to the variability of disabilities among individuals, the speech variations among those with dysarthria far exceed than healthy speech. Standard ASR struggles to effectively model these variations, failing to accurately map dysarthric speech to phonemes. ASR must confront the

challenges arising from inaccurate phonation, tempo, and inconsistencies in formant transitions to address the needs of dysarthric individuals [3-5]. Therefore, there is a growing need for specialized DSR systems designed specifically to accommodate the distinctive speech patterns of individuals with dysarthria. While some efforts have been made to implement ASR technologies for dysarthric speech, previous research indicates that ASR systems tailored for users with dysarthria have not achieved satisfactory accuracy levels due to the intricacies involved in DSR [6-9]. Factors such as heightened variability stemming from physical fatigue and frustration experienced by individuals with dysarthria, alongside variations in disease severity levels, pose challenges in developing an accurate ASR system suitable for individuals with dysarthria. Furthermore, recent research indicates that deep learning-based Voice Conversion (VC) methods have demonstrated enhanced speech intelligibility for individuals with dysarthria compared to conventional algorithms. For instance, Chen et al. introduced a deep learning architecture aimed at transforming dysarthric speech utilizing log-power spectra and MFCC [10]. In recent years, AI-based speech recognition systems have undergone significant evolution, demonstrating remarkable accuracy and efficiency across diverse applications. These systems play a pivotal role in popular virtual assistants such as Siri, Google Assistant, and Alexa, facilitating seamless interaction between users and devices through natural language commands. Particularly noteworthy among these advancements is the emergence of end-to-end models, representing a significant leap forward in ASR technology [11]. Unlike traditional speech recognition systems, which involve multiple stages including feature extraction, acoustic modeling, pronunciation modeling, language modeling, and decoding, end-to-end speech recognition aims to streamline this process by directly mapping input audio to output text without relying on intermediate representations.

II. RELATED WORKS

End-to-end speech recognition refers to a paradigm in ASR where a single model directly maps the acoustic features of an audio signal to the corresponding textual representation. This approach holds promise for comprehending dysarthric speech, boasting benefits like simpler architecture, lower computational demands, and potentially better performance. Dysarthric speech is characterized by imprecise articulation, reduced intelligibility, and irregular speech patterns due to weakness or paralysis of the muscles involved in speech production. Traditional ASR systems often struggle to accurately transcribe dysarthric speech due to its atypical acoustic properties and variability [12-14]. End-to-end speech recognition models have the potential to mitigate these

challenges by learning directly from the acoustic input to the textual output without relying on explicit linguistic or phonetic knowledge [15]. These models can capture complex patterns and variations present in dysarthric speech more effectively, enabling better transcription accuracy.

By leveraging deep learning architectures such as Recurrent Neural Networks (RNNs) or transformers, end-to-end models can adapt to the unique characteristics of dysarthric speech, including irregular prosody, inconsistent articulation, and variable speech rate [16]. Furthermore, end-to-end speech recognition can benefit from transfer learning techniques, where models pre-trained on large-scale general speech corpora can be fine-tuned on smaller datasets specific to dysarthric speech [17]. This process allows the model to leverage knowledge gained from abundant data sources to improve performance on tasks with limited training data. To optimize performance across diverse facets of dysarthric speech, a proposed strategy involves adopting a multi-learner approach [18]. This method entails training several Artificial Neural Networks (ANNs) separately to enhance different aspects, thereby collectively improving overall performance. RNNs and transformer models are capable of directly associating speech segments with character sequences [19], presenting a multitude of advantages. These include the integration of language understanding within the model and the potential for simultaneous optimization of all components, which could lead to enhanced performance. Nonetheless, DSR encounters distinctive challenges due to the scarcity of data and the intricate characteristics of dysarthric speech. Although RNN-based models have shown promise, especially in situations with scarce data, they frequently face challenges in capturing long-range dependencies [20]. Similarly, transformer-based models, despite their notable accomplishments in traditional speech recognition tasks, have not yet fully realized their potential in the domain of DSR [21, 22].

The latest trend in end-to-end ASR models is exemplified by Whisper, an innovative system trained on a vast corpus of 680,000 hours of multilingual and multitask supervised data sourced from the web [23, 24]. This comprehensive dataset contributes to enhancing the system's resilience to accents, background noise, and technical language intricacies. The Whisper model, rooted in transformer architectures, holds promise for addressing the unique challenges of DSR. It leverages the inherent capabilities of transformers, which excel in capturing long-range dependencies in sequential data. This enables the model to exhibit the potential for enhanced adaptability to the distinctive characteristics of dysarthric speech. The Whisper model's architecture enables the direct association of speech segments with character sequences, aligning with the fundamental principles of end-to-end DSR. It effectively maps acoustic features to text, enhancing transcription accuracy for dysarthric speech without relying on explicit linguistic or phonetic knowledge. Pre-training the model on large-scale general speech corpora and fine-tuning it on smaller datasets specific to dysarthric speech allows the Whisper model to leverage knowledge gained from abundant data sources, thereby enhancing its performance on tasks with limited training data availability.

III. SPEECH CORPORA

UA-Speech stands out as the largest dataset featuring a significant number of dysarthric participants and has been

extensively utilized in DSR research [25]. Developed by researchers at the University of Illinois, UA-Speech offers a comprehensive range of speech tasks, including sustained vowel phonation and syllable repetition, allowing for diverse analyses of speech production. The recordings, conducted in controlled acoustic environments using high-quality microphones, are meticulously annotated with details regarding speech intelligibility, prosody, and phonetic errors. This extensive collection encompasses recordings from 19 individuals with dysarthria, exhibiting varying levels of speech intelligibility ranging from 2% to 95%. These levels are categorized from very low (0-25%) to low (25-50%), mild (50-75%), and high (75-100%) intelligibility. Addressing the difficulty of incorporating the extensive UA Speech database, which comprises 66,280 speech signals, into models, steps have been implemented to enhance accessibility. The dataset has been methodically organized and made accessible on the Hugging Face platform, complete with train and test splits. Furthermore, access to the dataset is contingent upon approval, ensuring responsible usage and adherence to ethical guidelines. The proposed model adopts a 75:25 train/test split for UA-Speech, ensuring a balanced and representative distribution of training and evaluation datasets.

IV. PROPOSED WHISPER MODEL FOR DSR

The Whisper architecture represents a significant advancement in the field of speech recognition through its innovative design and utilization of state-of-the-art techniques. The model employs an end-to-end approach, directly mapping input audio signals to output text captions without relying on intermediate representations or separate processing stages. At the heart of the Whisper architecture lies an encoder-decoder Transformer model, which captures long-range dependencies and contextual information effectively, enabling the model to learn complex mappings between input audio features and corresponding textual transcriptions. Before being fed into the encoder, the input audio is segmented into 30-second chunks and then converted into log-mel spectrograms to capture frequency content over time, essential for feature extraction. The encoder processes these spectrogram representations, extracting high-level features and capturing important acoustic characteristics of the speech signal. These features are then passed to the decoder, responsible for generating textual transcriptions. Special tokens within the decoder facilitate tasks such as language identification, phrase-level timestamps, multilingual speech transcription, and to-English speech translation, expanding the model's versatility and applicability. Moreover, the Whisper model's robust error-handling mechanisms make it particularly well-suited for fine-tuning with dysarthric speech. By accommodating variations in speech quality and environmental noise, the model can adapt more effectively to the unique challenges posed by dysarthria. This adaptability allows for targeted adjustments during fine-tuning processes, optimizing the model's performance to better capture the distinctive speech patterns and nuances characteristic of dysarthric individuals. Additionally, the Whisper model's resilience to such variations ensures that fine-tuning efforts remain effective across a range of real-world scenarios, ultimately leading to more accurate and reliable speech recognition outcomes for dysarthric users.

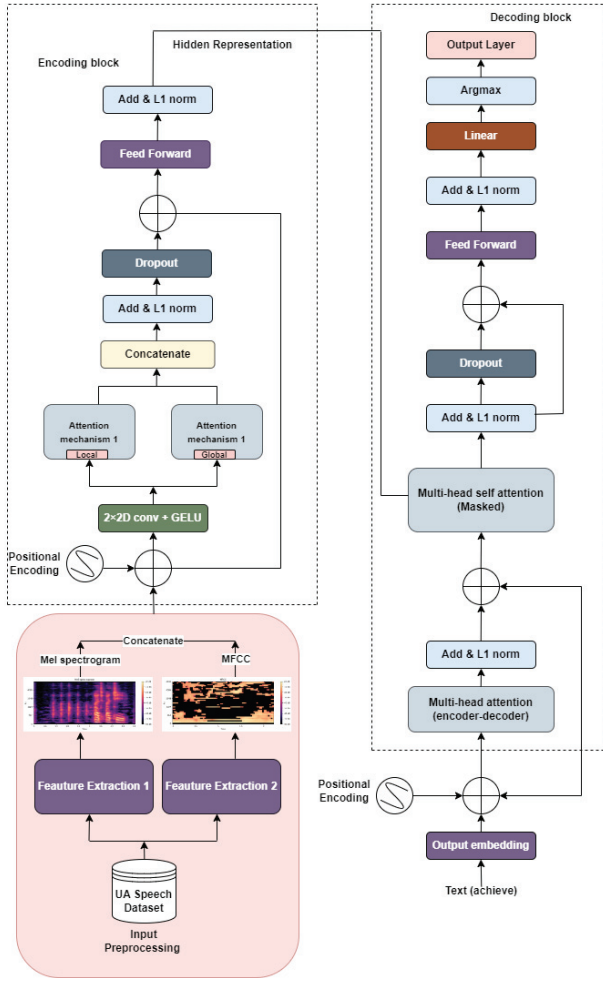


Fig. 1. Proposed DSR Model

Fine-tuning the proposed Whisper model for DSR involves building upon its existing framework, which primarily relies on spectrograms for feature extraction. However, to address the challenges posed by dysarthric speech characteristics such as imprecise articulation and slower speaking rates, the model incorporates additional features extracted from MFCCs as shown in Fig. 1. By combining spectrograms and MFCCs within an attention mechanism, the model can create a richer feature representation, with spectrograms offering broader context and MFCCs highlighting crucial details like formant frequencies. Figures 2-5 exhibit few mel spectrogram and MFCC obtained during feature extraction process of proposed model. These examples specifically focus on the utterance of the word "achieve" and highlight speakers with varying degrees of speech intelligibility. The speakers featured include individuals identified as M04 (very low), M05 (mild), M07 (low), and M09 (high). The attention mechanism can be adapted to simultaneously attend to both modalities, allowing the model to weigh the importance of each feature dynamically based on specific speech segments and dysarthric speech characteristics. Additionally, a hierarchical attention approach is employed, involving a two-stage attention process focusing on local and global attention to capture both fine-grained details and broader contextual information within the speech signal. Hierarchical attention can improve the model's ability to handle dysarthric speech complexities, although it requires careful architecture design and may necessitate more computational resources for training.

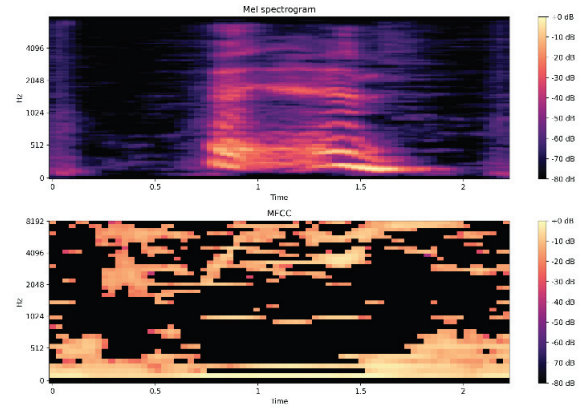


Fig. 2. Mel spectrogram and MFCC of Speaker M04 uttered word "achieve"

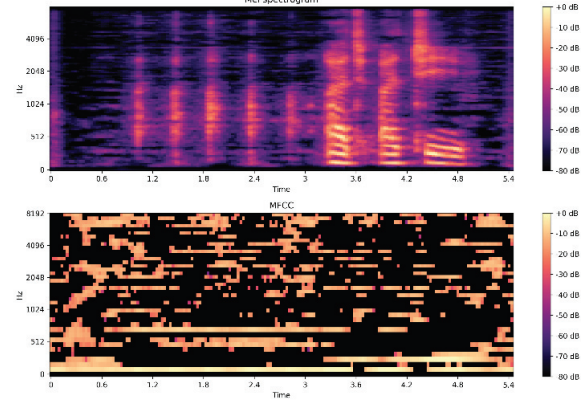


Fig. 3. Mel spectrogram and MFCC of Speaker M05 uttered word "achieve"

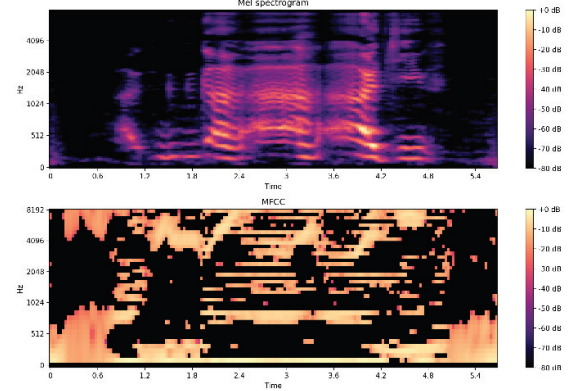


Fig. 4. Mel spectrogram and MFCC of Speaker M07 uttered word "achieve"

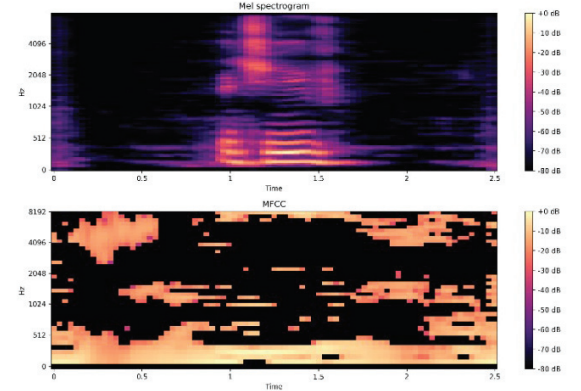


Fig. 5. Mel spectrogram and MFCC of Speaker M09 uttered word "achieve"

V. RESULTS AND DISCUSSIONS

All experiments were conducted on a specialized deep learning workstation, a Lenovo Think Station equipped with an Intel Xeon processor, 128GB RAM, and four NVIDIA Quadra RTX 8000 GPUs. Throughout the training of the proposed model, a learning rate of 0.0003 is maintained over 15 epochs, with a batch size of 32. The performance metrics are depicted in Figures 7 to 9. Fig 7 illustrates the decreasing trend of both training and evaluation loss per epoch, reflecting the model's effective learning and generalization capabilities. In Fig 8, the stable evaluation samples per second suggest efficient computation during evaluation, despite fluctuations in loss. Furthermore, Fig 9 showcases the model's improving accuracy over epochs, as evidenced by the declining evaluation loss. However, a slight increase in runtime per epoch, as observed in Figure 8, indicates the trade-off between computational complexity and performance optimization. Table II establishes a detailed comparison between the WRA obtained from the benchmark and the proposed DSR model across various speakers and levels of speech intelligibility. The performance analysis comparing the proposed model to the benchmark system is illustrated in Fig 6. In instances of "High" intelligibility, the proposed DSR model either maintains or slightly enhances the WRA percentages compared to the benchmark. Notably, speakers F05, M09, M10, M14, and M08 all attain WRA percentages exceeding 85%, with speaker M09 notably improving from 84% in the benchmark to 90% with the proposed model. Progressing to the "Medium" intelligibility level, the proposed DSR model exhibits significant improvements for speakers M11, M05, and F04, all showcasing enhanced WRA percentages compared to the benchmark. Notably, speaker F04 demonstrates a substantial improvement, with the WRA increasing from 72% to 76%. At the "Low" intelligibility level, the proposed DSR model maintains comparable performance to the benchmark for speaker M07, achieving identical WRA percentages of 76%. However, for speaker F02, the proposed model achieves a higher WRA of 78%, representing a notable improvement from the benchmark's 74%. In cases of "Very Low" intelligibility, the proposed DSR model markedly outperforms the benchmark for speakers F03, M04, and M12. Particularly noteworthy is the significant improvement observed in speaker M04, with the WRA soaring from 9% in the benchmark to 65% with the proposed model. These results underscore the effectiveness of the proposed DSR model in enhancing ASR accuracy across a spectrum of speech intelligibility levels, especially for speakers with severe speech impairments.

TABLE I. COMPARATIVE ANALYSIS OF WRA BASED ON UA DYSARTHIC SPEECH CORPUS

Speaker ID	Intelligibility Level	WRA % [17] Benchmark	WRA% Proposed DSR model
F05	High	93%	93%
M09	High	84%	90%
M10	High	91%	89%

M14	High	85%	88%
M08	High	88%	87%
M11	Medium	58%	69%
M05	Medium	66%	71%
F04	Medium	72%	76%
F02	Low	74%	78%
M07	Low	76%	74%
F03	Very Low	51%	71%
M04	Very Low	9%	65%
M12	Very Low	53%	62%

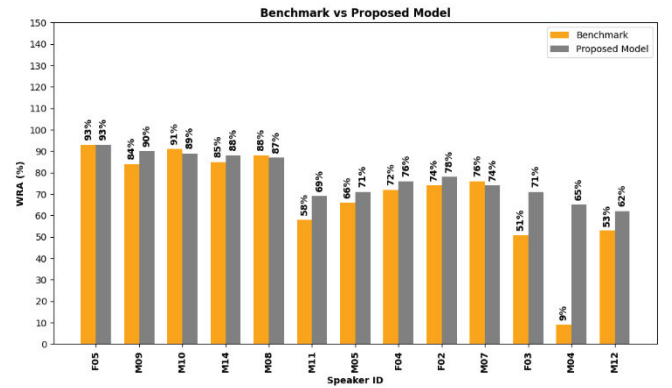


Fig. 6. Performance of benchmark vs proposed model

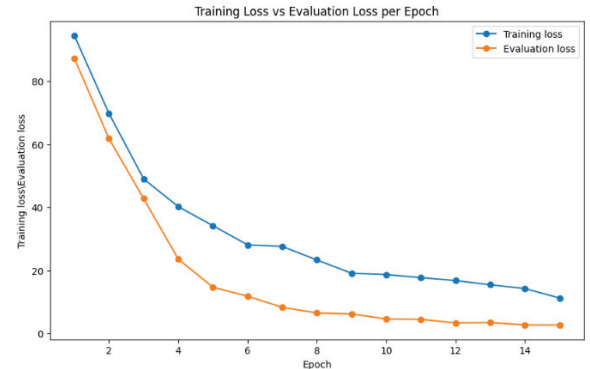


Fig. 7. Training and evaluation loss per epoch

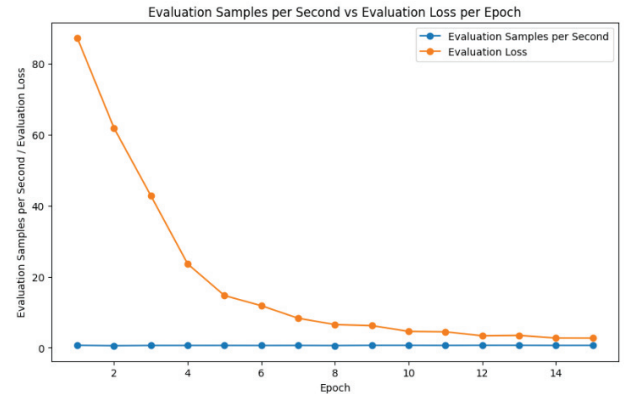


Fig. 8. Evaluation samples per second vs Evaluation loss per epoch

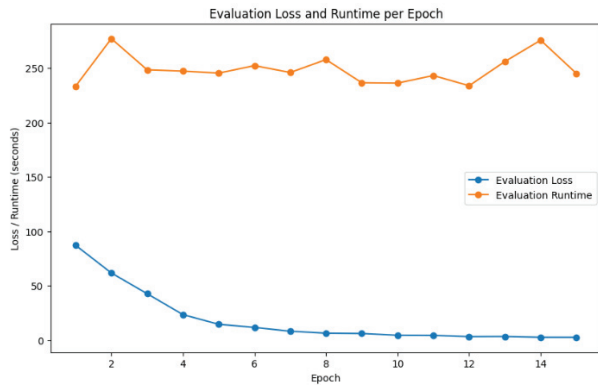


Fig. 9 Evaluation loss and runtime per epoch

VI. CONCLUSION

In this study, the Whisper model has been utilized for DSR, recognizing its potential to address the challenges faced by individuals with dysarthria. Dysarthric speech presents impaired patterns, posing obstacles for traditional ASR systems. To better understand and learn from these speech nuances, MFCC has been incorporated into the Whisper model. By combining spectrograms and MFCCs within an attention mechanism, our proposed model creates a richer feature representation, enabling it to capture fine-grained details and broader contextual information within the speech signal. Additionally, a hierarchical attention approach is employed, involving a two-stage attention process focusing on local and global attention to capture both fine-grained details and broader contextual information within the speech signal, varying levels of speech intelligibility. Experimental results reveal significant improvements in WRA achieved by the proposed DSR model, with a WRA of 74.08%, surpassing the benchmark DSR systems which achieved 69.23%. This enhancement is particularly noteworthy for speakers with severe speech impairments, as individuals with "Very Low" intelligibility levels witnessed substantial improvements in WRA. Overall, the findings emphasize the effectiveness of the proposed DSR model in enhancing ASR accuracy and accessibility for individuals with dysarthria.

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